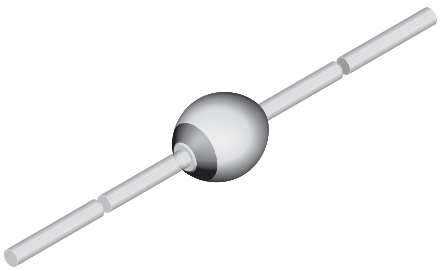
# Fast Avalanche Sinterglass Diode



949539

## FEATURES

- Glass passivated junction
- Hermetically sealed package
- Low reverse current
- Soft recovery characteristics
- Material categorization:
  For definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT
HALOGEN FREE

## APPLICATIONS

- Fast “soft recovery” rectification diode

## MECHANICAL DATA

**Case:** SOD-57

**Terminals:** plated axial leads, solderable per MIL-STD-750, method 2026

**Polarity:** color band denotes cathode end

**Mounting position:** any

**Weight:** approx. 369 mg

**ORDERING INFORMATION** (Example)

| DEVICE NAME | ORDERING CODE | TAPED UNITS | MINIMUM ORDER QUANTITY |
|---|---|---|---|
| BYV38 | BYV38-TR | 5000 per 10" tape and reel | 25 000 |
| BYV38 | BYV38-TAP | 5000 per ammopack | 25 000 |

**PARTS TABLE**

| PART | TYPE DIFFERENTIATION | PACKAGE |
|---|---|---|
| BYV37 | $V_R$ = 800 V; $I_{F(AV)}$ = 2 A | SOD-57 |
| BYV38 | $V_R$ = 1000 V; $I_{F(AV)}$ = 2 A | SOD-57 |

**ABSOLUTE MAXIMUM RATINGS** ($T_{amb}$ = 25 °C, unless otherwise specified)

| PARAMETER | TEST CONDITION | PART | SYMBOL | VALUE | UNIT |
|---|---|---|---|---|---|
| Reverse voltage | See electrical characteristics | BYV37 | $V_R = V_{RRM}$ | 800 | V |
| | | BYV38 | $V_R = V_{RRM}$ | 1000 | V |
| Peak forward surge current | $t_p$ = 10 ms, half sine wave | | $I_{FSM}$ | 50 | A |
| Average forward current | | | $I_{F(AV)}$ | 2 | A |
| Non repetitive reverse avalanche energy | $I_{(BR)R}$ = 0.4 A | | $E_R$ | 10 | mJ |
| Junction and storage temperature range | | | $T_j = T_{stg}$ | - 55 to + 175 | °C |

**MAXIMUM THERMAL RESISTANCE** ($T_{amb}$ = 25 °C, unless otherwise specified)

| PARAMETER | TEST CONDITION | SYMBOL | VALUE | UNIT |
|---|---|---|---|---|
| Junction ambient | Lead length l = 10 mm, $T_L$ = constant | $R_{thJA}$ | 45 | K/W |
| | On PC board with spacing 25 mm | $R_{thJA}$ | 100 | K/W |



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| **ELECTRICAL CHARACTERISTICS** ($T_{amb}$ = 25 °C, unless otherwise specified) | | | | | | | |
|---|---|---|---|---|---|---|---|
| **PARAMETER** | **TEST CONDITION** | **PART** | **SYMBOL** | **MIN.** | **TYP.** | **MAX.** | **UNIT** |
| Forward voltage | $I_F$ = 1 A | | $V_F$ | - | 1 | 1.1 | V |
| Reverse current | $V_R = V_{RRM}$ | | $I_R$ | - | - | 5 | µA |
| | $V_R = V_{RRM}$, $T_j$ = 150 °C | | $I_R$ | - | - | 150 | µA |
| Reverse recovery time | $I_F$ = 0.5 A, $I_R$ = 1 A, $i_R$ = 0.25 A | | $t_{rr}$ | - | - | 300 | ns |
| Diode capacitance | $V_R$ = 4 V, f = 1 MHz | | $C_D$ | - | 15 | - | pF |

## TYPICAL CHARACTERISTICS ($T_{amb}$ = 25 °C, unless otherwise specified)

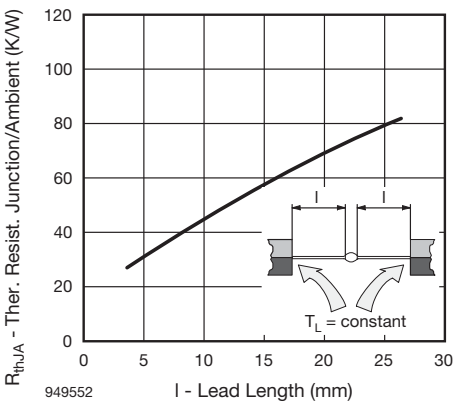



Fig. 1 - Max. Thermal Resistance vs. Lead Length

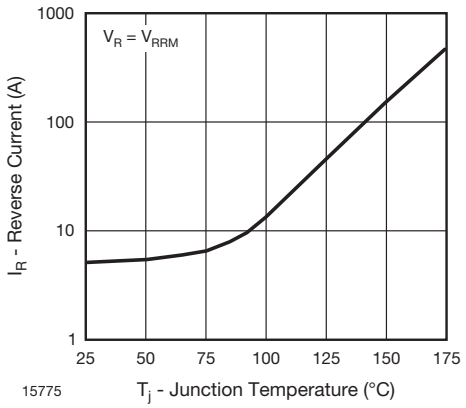



Fig. 3 - Max. Reverse Current vs. Junction Temperature

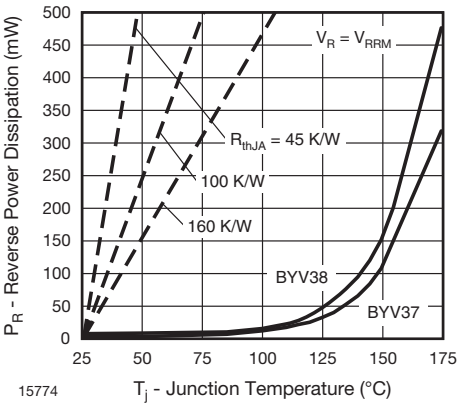



Fig. 2 - Max. Reverse Power Dissipation vs. Junction Temperature

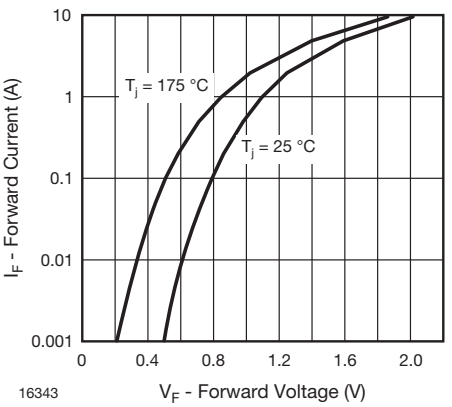



Fig. 4 - Forward Current vs. Forward Voltage



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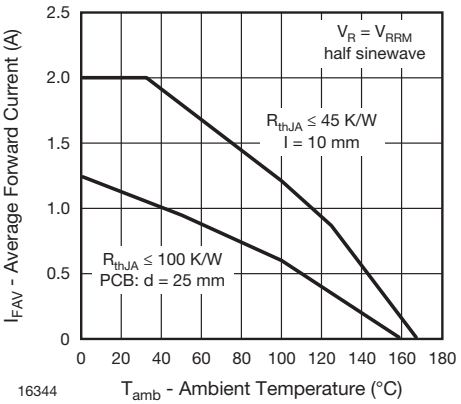



Fig. 5 - Max. Average Forward Current vs. Ambient Temperature

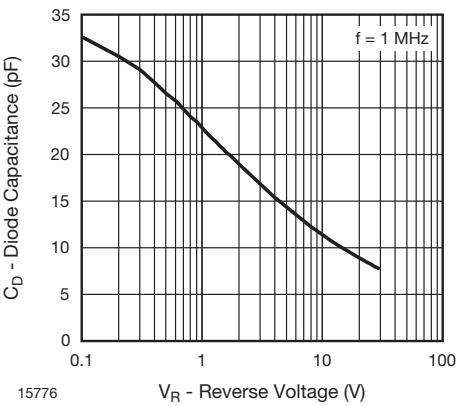
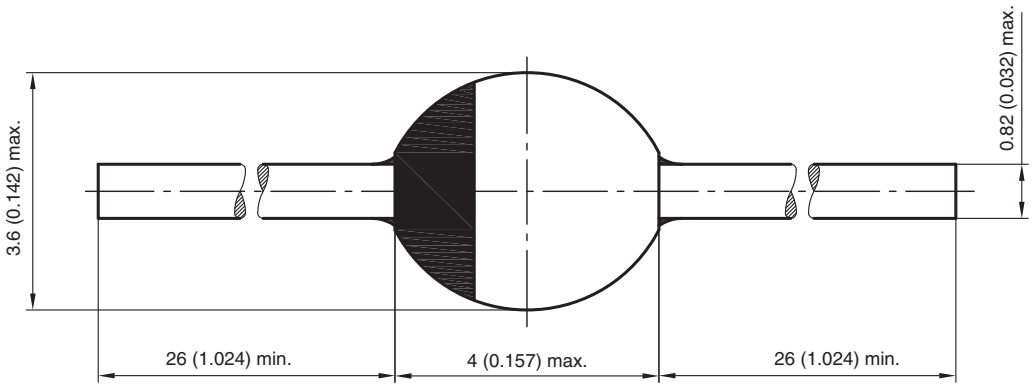



Fig. 6 - Typ. Diode Capacitance vs. Reverse Voltage

**PACKAGE DIMENSIONS** in millimeters (inches): **SOD-57**




20543
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